



ENGIN 604 INTRODUCTION TO PYTHON FOR FINANCE — FALL 2021

HOMEWORK 4

Due: 11:59pm, Saturday, April 17

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Maximum page limit: 5 pages

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The file `ETFs.xlsx` contains the OHCL (*Open, High, Close and Low*) of the following *Exchange Traded Funds*¹ at a daily frequency:

- SPY** (S&P500 ETF), from 2012-11-26 to 2020-12-30
- EFA** (a non-US equities ETF), from 2012-12-31 to 2020-12-30
- IJS** (a small-cap value ETF), from 2012-10-31 to 2020-12-30
- EEM** (an emerging-markets ETF), from 2012-04-25 to 2020-12-30
- AGG** (A bond ETF), from 2012-12-31 to 2020-12-30

All the data were extracted from *Yahoo Finance* using the `YFinance` library.

- Load the file `ETFs.xlsx` into your workspace so that each sheet is stored in a dictionary.
- Build a `DataFrame` containing the closing price of each ETF.
 - The column (`columns`) must have the name of the ETF and the index (`index`) the date.
 - Work only with the data from 2012-12-31 to 2020-12-30.

Explain at least two ways of doing (2.b).

- Transform the `DataFrame` created in (2) to a monthly frequency. This is done by assuming that the last available price of the month corresponds to the monthly price.
- Using a `lambda` (anonymous) function, generate the monthly logarithmic return for each ETF. Do not forget to drop the row of `NAs`.

$$R_{j,t} = \ln \left(\frac{P_{j,t}}{P_{j,t-1}} \right) \quad j = 1, \dots, 5 \quad t = 1, \dots, 97$$

Hint: NumPy has the `np.log()` function to compute the natural logarithm.

- Write a function that computes a brief descriptive statistic containing: arithmetic mean, standard deviation, *kurtosis*, *skewness* and Sharpe ratio (assume a risk-free rate equal to 0). For the function, take the following into account:
 - The *input* must be a Pandas with a Series structure.
 - The *output* must be a Pandas with a Series structure where:

¹These are collective investment funds whose investment policy consists of replicating an index.

- The `name` attribute contains the name of the *input*.
- The `index` attribute contains the name of each descriptive statistic computed.

Apply the function to the DataFrame created in (4) by means of a `lambda` (anonymous) function.

- Using a line chart, plot the cumulative logarithmic return of each ETF.
- The tangency portfolio is the one that maximizes the Sharpe ratio. The optimization problem expressed in its matrix form is:

$$\max_{\mathbf{t}} \frac{\mathbf{t}'\boldsymbol{\mu} - r_f}{(\mathbf{t}'\boldsymbol{\Sigma}\mathbf{t})^{\frac{1}{2}}} = \frac{\mu_{p,t} - r_f}{\sigma_{p,t}} \quad s.t \quad \mathbf{t}'\mathbf{1} = 1$$

Whose solution (also in matrix form) is:

$$\mathbf{t} = \frac{\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f \cdot \mathbf{1})}{\mathbf{1}'\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f \cdot \mathbf{1})}$$

Where $\mathbf{t}$ is the vector of optimal weights of the tangency portfolio, $\mathbf{1}$ a vector of ones whose dimension is $j \times 1$, with j the total number of assets, and $\boldsymbol{\Sigma}^{-1}$ the inverse of the variance-covariance matrix ($j \times j$).

Using the matrix solution for $\mathbf{t}$, write a function whose *inputs* are: the vector of average returns, the variance-covariance matrix of the assets (ETFs) that would make up the tangency portfolio and the risk-free rate (scalar).

The *output* of the function must be the weights $\mathbf{t}$, the expected return and the standard deviation of the tangency portfolio.

- The expected return of the tangency portfolio is:

$$R_{p,t} = \mathbf{t}' \cdot \boldsymbol{\mu}$$

Where $\boldsymbol{\mu}$ is the vector of average returns of dimension $j \times 1$.

- The standard deviation of the minimum-variance portfolio is:

$$\sigma_{p,t} = \sqrt{\mathbf{t}'\boldsymbol{\Sigma}\mathbf{t}}$$

Test the function using the average returns created in (5) and build the variance-covariance matrix using the DataFrame generated in (4). Explain your function line by line. Assume a risk-free rate equal to 0.005.

Hint: To convert rows or columns of a DataFrame to NumPy, use the `.to_numpy()` attribute after the selection (`loc` or `iloc`).